Quinten Pienaar | quinten.pienaar@gmail.com | https://github.com/qpienaar/school_projects/tree/main | http://www.linkedin.com/in/qpienaar |

Education

Rensselaer Polytechnic Institute, Troy, NY
Bachelor's of Science in Mechanical Engineering, Minor in Economics, GPA 3.53, Cum Laude — Fall 2022-Fall 2025
Master's of Science in Mechanical Engineering, GPA 3.92 — Spring 2026-Fall 2026

Experience

Academic Research, Center for Scientific Computation, Troy, NY — Summer 2025-Present
- Conducting research into performance and accuracy differences between Constructive Solid Geometry (CSG) based particle transport simulations and Boundary Representation (BREP) mesh based particle transport simulations
- Developing a robust utility to convert from CSG based 3d models to BREP based 3d models
- Quantifying required mesh granularity and computation power needed for accurate GPU accelerated mesh based particle transport simulations

Product Security Intern, GE Vernova - Wind, Schenectady, NY — Summer 2025
- Proposed segmentation solutions to mitigate unauthorized movement within industrial control network to senior management
- Investigated micropatching strategies to address critical software vulnerabilities using DARPA tool for embedded software
- Designed role-based cybersecurity training program tailored to internal staff across multiple job functions
- Tested and recommended internal tools to enhance secure coding practices within development teams

Control Systems Intern, GE Vernova - Gas Power, Schenectady, NY — Summer 2024
- Worked in the GEV's Gas Power Controls lab
- Tested and qualified digital bus devices for customer use
- Developed controls application software and integrated lab hardware
  - Mark Vle, Woodward valve actuator, HMI station
- Simulated customer control environments for testing purposes
- Supported new networking device introduction
- Contributed to lab safety continuous improvement

Projects

Robotic Path Planning with Neural Networks - Scientific Machine Learning Final Project, Rensselaer Polytechnic Institute — Spring 2026
- Trained two convolutional neural networks, one with a mean squared error loss function, and one with an asymmetric piecewise mean absolute error loss function, to estimate the distance from a point to a goal in a 2d environment with obstacles
- Generated and conducted an experiment to evaluate neural networks ability to reliably improve upon euclidean distance as a heuristic function for robotic navigation in 2d environments with varying types of obstacles
- Created and gave a technical presentation of work and findings to peers and faculty

Cable-in-Conduit Thermal Analysis - Introduction to Computational Fluid Dynamics, Rensselaer Polytechnic Institute — Spring 2026
- Developed 3d model of a representative Cable-in-Conduit section, used for cooling of magnetic strands in Tokamak fusion reactors
- Applied volume, surface, and boundary layer FEM/FEA meshes to 3d model and applied fluid/thermal boundary conditions
- Conducted a high fidelity CFD simulation and analysis to understand conjugate thermal/fluid movement through cooling systems

Parallel Image Recreation and Rendering - Parallel Computing Final Project, Rensselaer Polytechnic Institute — Spring 2026
- Collaborated in a small group of graduate students to implement a massively parallel, GPU accelerated, genetic algorithm for image reconstruction
- Implemented code base on RPI's AiMOS supercomputing cluster and conducted strong and weak scaling experiments in single node multi-GPU, and multi node multi-GPU MPI approaches
- Synthesized data and developed a conference-ready paper of our approach and findings

Moving Target Visibility Graph - Robotics II, Rensselaer Polytechnic Institute — Spring 2026
- Developed a Moving Target Visibility Graph (MTVG) algorithm to plan interception paths for an agent pursuing a moving target with obstacles
- Implemented geometric visibility, time-dependent interception feasibility, and edge-cost calculations using an object-oriented Python approach
- Conducted strong scaling studies against static visibility graph construction to understand computational cost and scaling

Wing Spar Geometry Optimization Under Uncertainty - Numerical Design Optimization Project, Rensselaer Polytechnic Institute — Fall 2025
- Optimized a carbon-fiber UAV wing spar under uncertain aerodynamic loading using MATLAB fmincon, minimizing structural mass while enforcing six-sigma stress constraints
- Achieved a 70.68% reduction in spar mass relative to the nominal design while satisfying probabilistic structural performance requirements

Skills

Languages & Tools: Python, MATLAB, Simulink, C++, C, CUDA, PyTorch, Matplotlib, MPI
Engineering: FEM/FEA, CFD, Particle Transport (CSG/BREP), HPC & GPU Acceleration, Numerical Optimization
Other: Scientific ML, Dynamic Systems Modeling, Feedback Control, State-Space Analysis, LQR Control, Surrogate Modeling

Leadership and Activities

- Pi Lambda Phi Fraternity - President Fall 2024 - Fall 2025
  - Oversaw ~$20k facility upgrades, reformed minor position standards.
- Rensselaer Men's Rugby - Captain Spring 2025 - Fall 2025
  - Led team to National 7s playoff appearance, restructured practices for more efficient time management.
- Rensselaer Club Men's Basketball - Vice President Fall 2024 - Spring 2025
  - Led team to regional semifinal appearance, created club constitution outlining division of labor and expectations.